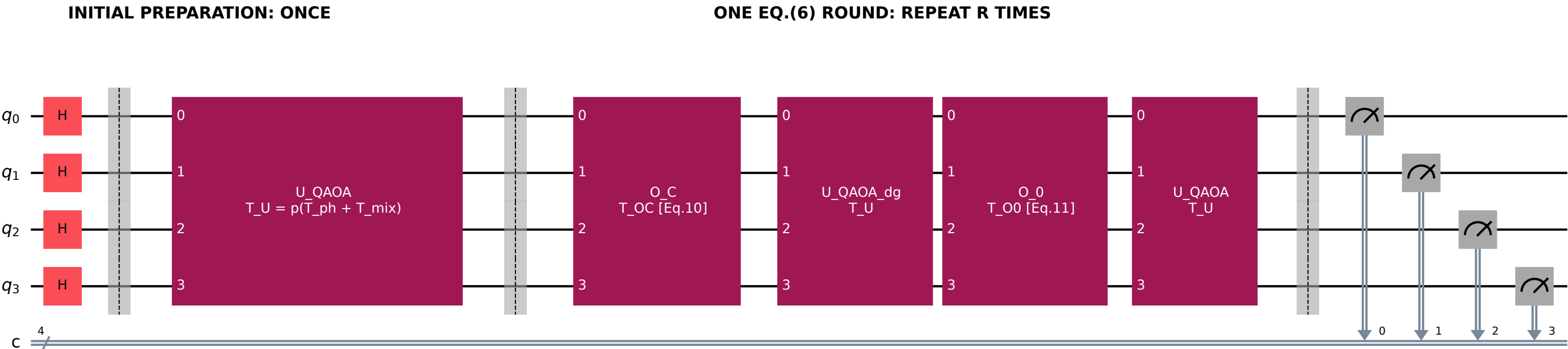


QAOA preparation + literal paper Eq.(6): time flows left to right

Omanakuttan et al., arXiv:2504.01897v1 (2025) | logical-circuit view; time units: T_{LC} = d x 1 microsecond



Four drawn data wires represent an arbitrary n-qubit register. Boxes are schematic, not executable.

Initial H -> U_{QAOA} prepares $|\psi\rangle$. Repeat $O_C \rightarrow U_{dg} \rightarrow O_0 \rightarrow U$ exactly R times.

Each AA round: $T_Q = 2p(T_{phaser} + T_{mixer}) + T_{OC} + T_{O0}$. Eq.(7): $T_q = R T_Q$.

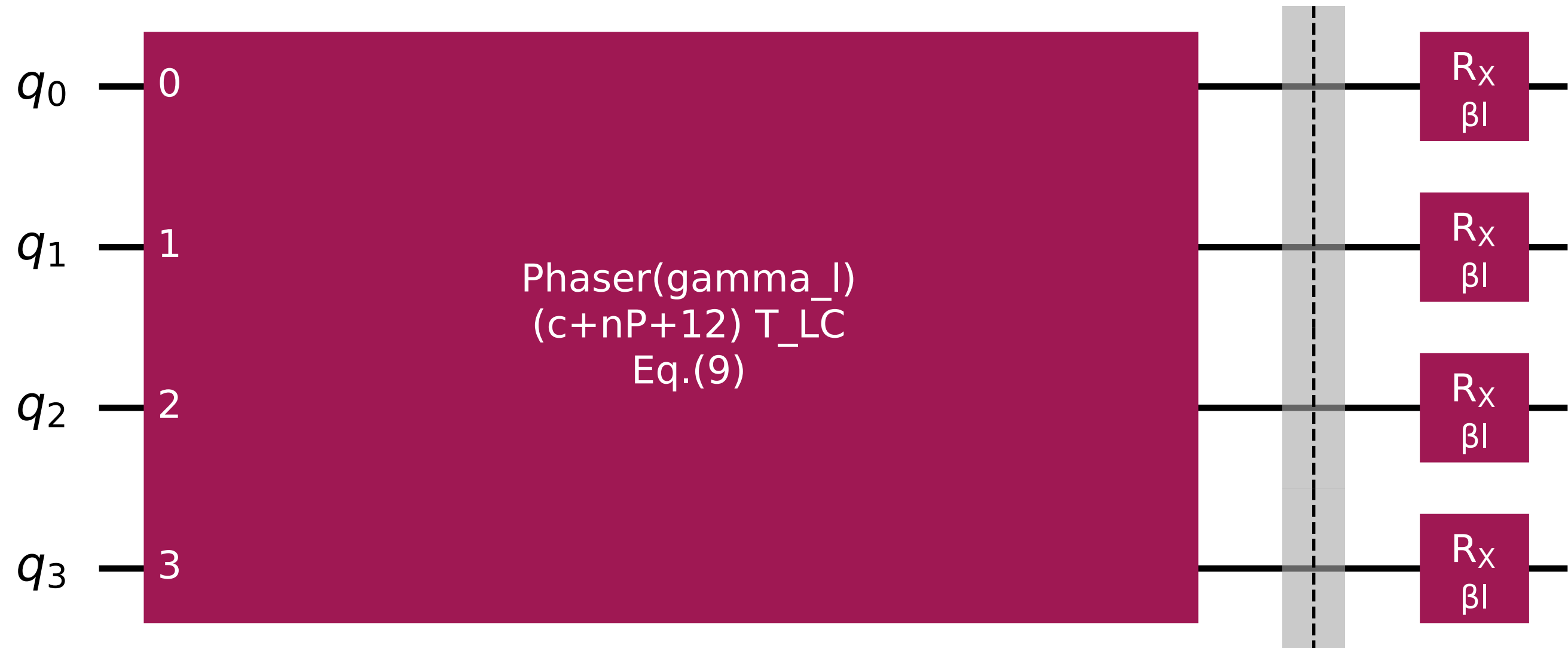
Literal Eq.(6), no outer H; preparation convention differs from Eqs.(2)-(3). See reading audit.

Initial preparation adds T_U (plus unspecified H/init costs). Final measurement/verification costs are unspecified.

Arrows/gates: left-to-right execution. Resource formulas belong to the paper compiler, not Qiskit circuit.depth().

One forward QAOA layer: phaser first, mixer second

Omanakuttan et al., arXiv:2504.01897v1 (2025) | logical-circuit view; time units: $T_{LC} = d \times 1 \text{ microsecond}$



Layer l : $\exp(-i \beta_l H_M) \exp(-i \gamma_l H_C)$, Eqs.(3)-(5); repeat $l=1, \dots, p$.

Parallel mixer: $R_x(\beta_l)$ on ALL n wires. $T_{\text{mixer}} = nP T_{LC}$ [Eq.(8)], not $n \cdot nP \cdot T_{LC}$.

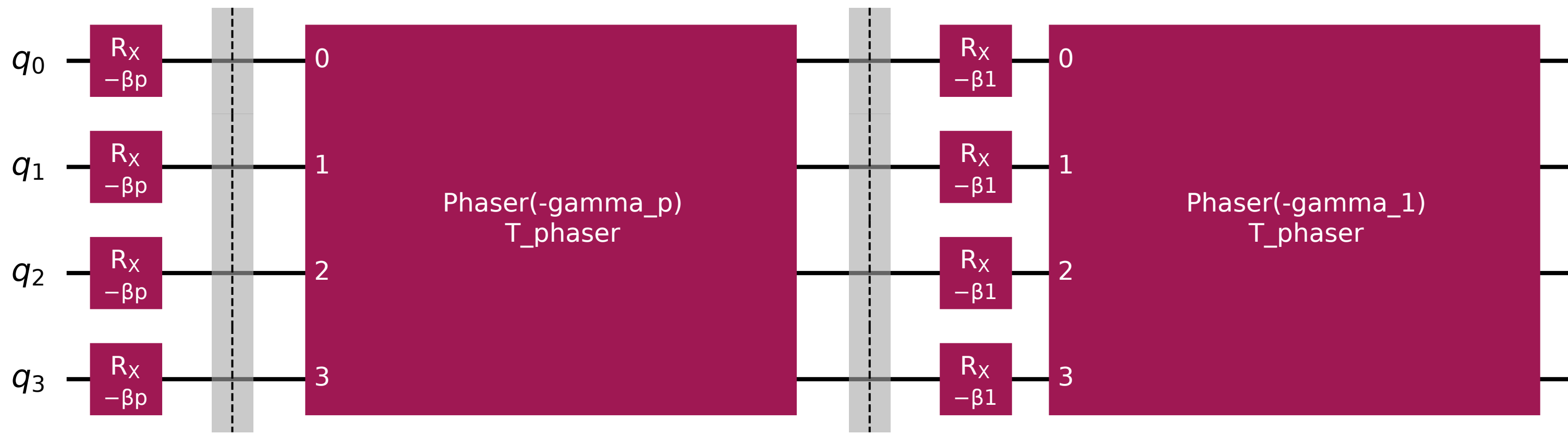
Whole phaser: $T_{\text{phaser}} = (c+nP+4 \text{ ceil}(\log_2 k)) T_{LC}$ [Eq.(9)]. $k=8$ gives $+12$.

One U_{QAOA} costs $p(T_{\text{phaser}}+T_{\text{mixer}})$. One U_{dg} costs the same in this resource model.

Arrows/gates: left-to-right execution. Resource formulas belong to the paper compiler, not Qiskit circuit.depth().

Inverse QAOA: reverse layer order AND gate order

Omanakuttan et al., arXiv:2504.01897v1 (2025) | logical-circuit view; time units: $T_{LC} = d \times 1$ microsecond



Shown: the last inverse layer p , then the first inverse layer 1 ; intermediate layers omitted.

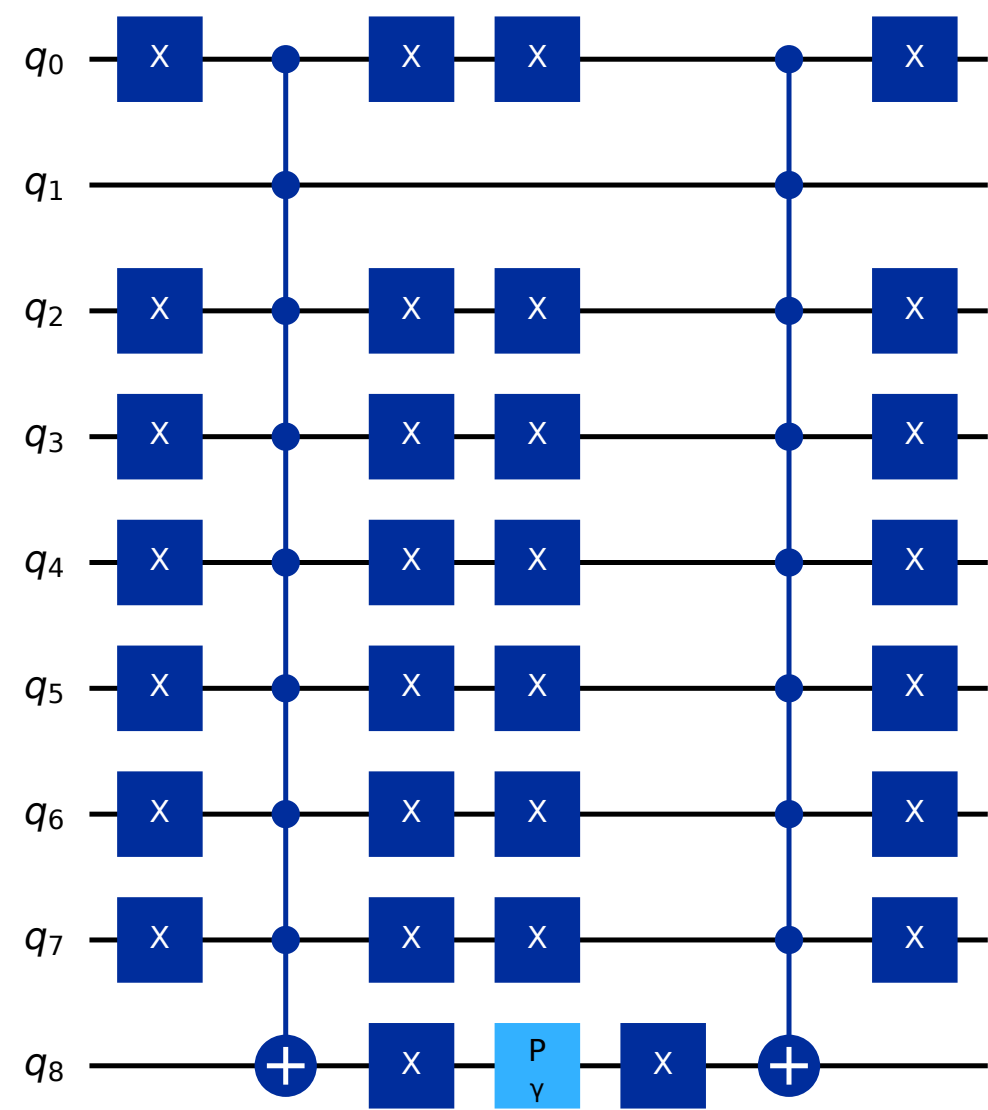
Forward layer: $\text{Phaser}(\gamma_l) \rightarrow \text{Mixer}(\beta_l)$. Inverse: $\text{Mixer}(-\beta_l) \rightarrow \text{Phaser}(-\gamma_l)$.

Run layers $p, p-1, \dots, 1$. Qiskit `.inverse()` performs both reversals and angle sign changes.

The inverse contributes another $p(T_{\text{phaser}} + T_{\text{mixer}})$ in every AA round [Eq.(7)].

Ideal 8-SAT phaser: positive phase follows Eqs.(3)-(4), unlike Eq.(20)

Omanakuttan et al., arXiv:2504.01897v1 (2025) | logical-circuit view; time units: T_{LC} = d x 1 microsecond



Clause: $x_0 \text{ OR } \text{NOT } x_1 \text{ OR } x_2 \text{ OR } \dots \text{ OR } x_7$. Positive literals receive X around the MCX.

P(+gamma) follows Eqs.(3)-(4); Eq.(20) prints the opposite sign. See the marked reading audit.

This MCX drawing explains semantics. It is NOT the optimized lattice-surgery circuit used to obtain Eq.(9).

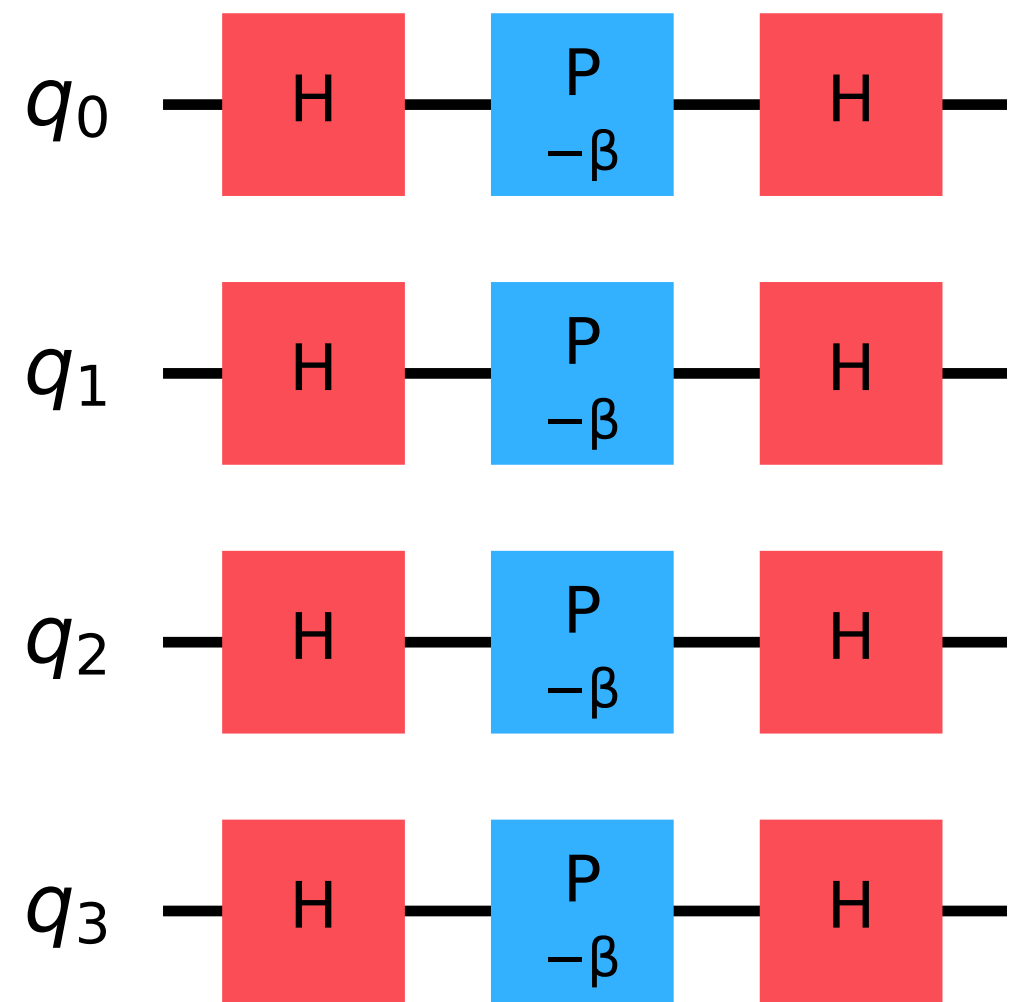
Paper timing: one TACU phase gadget runs $nP+4 \lceil \log_2 k \rceil$ cycles; $k=8$ uses 7 CCZ states and 52 ancillas.

All m clauses form c parallel batches; the FULL phaser time is $(c+nP+12) T_{LC}$, not m times this gadget.

Arrows/gates: left-to-right execution. Resource formulas belong to the paper compiler, not Qiskit circuit.depth().

Mixer as parallel phase gates in the X basis

Omanakuttan et al., arXiv:2504.01897v1 (2025) | logical-circuit view; time units: T_{LC} = d x 1 microsecond



$R_x(\beta) = \exp(-i \beta X/2)$ [paper Fig.1 caption; IBM RXGate definition].

$H P(-\beta) H = \exp(-i \beta/2) R_x(\beta)$; the difference is a global phase per wire.

All n phase decompositions run in parallel with sufficient T-state factories.

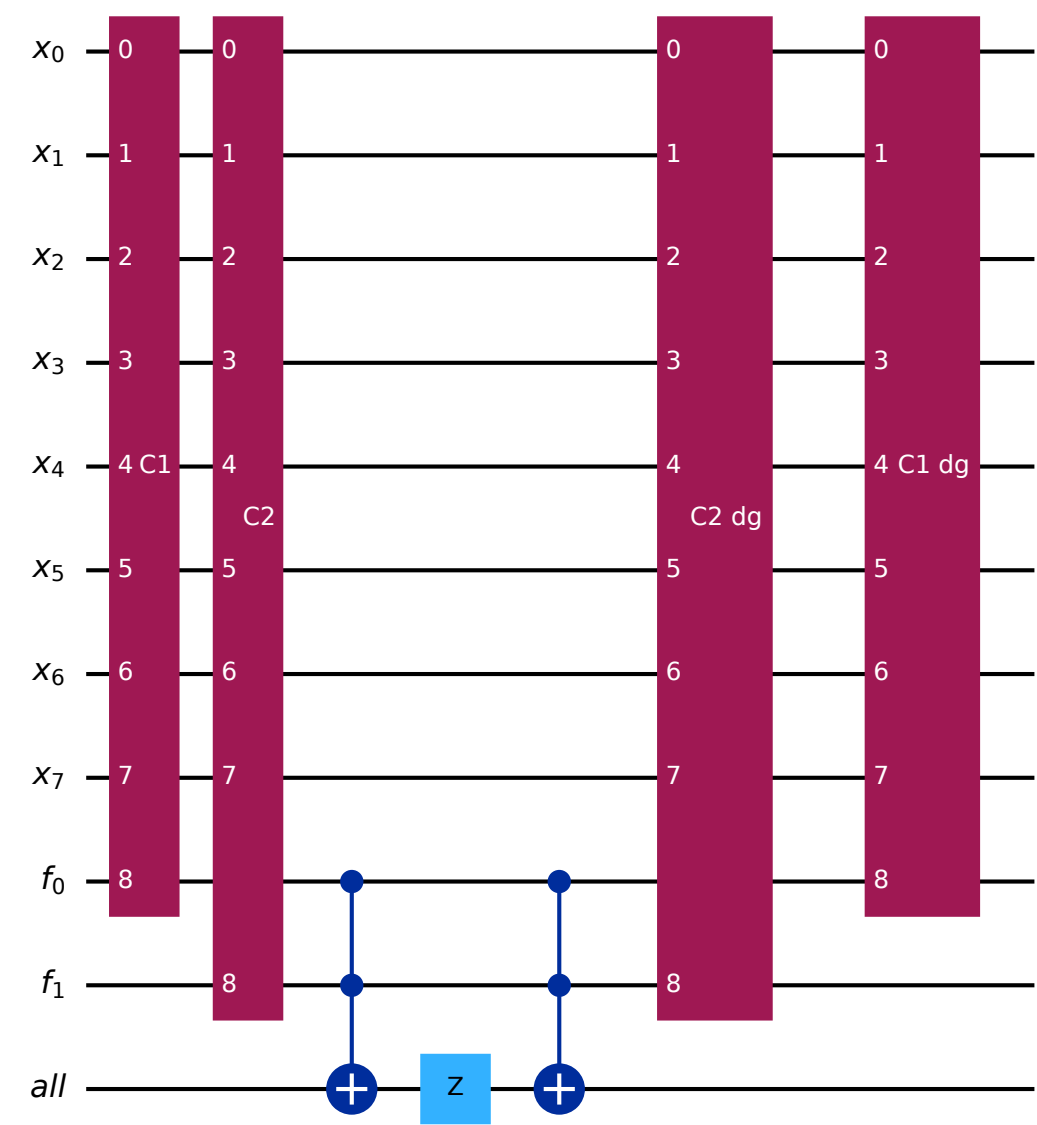
$T_{\text{mixer}} = nP T_{\text{LC}}$ [Eq.(8)]; nP is decomposition logical-cycle cost, not QAOA depth p .

Arrows/gates: left-to-right execution. Resource formulas belong to the paper compiler, not Qiskit circuit.depth().

SAT oracle: compute clauses, AND, Z, uncompute

Omanakuttan et al., arXiv:2504.01897v1 (2025) | logical-circuit view; time units: T_{LC} = d x 1 microsecond

FULL ORACLE: $4c \log_2(km/c) T_{LC}$ [Eq.(10)]



Executable illustration: n=8 variables, m=2 clauses, two clause flags and one all-clauses flag.

$O_C |x\rangle = (-1)^{(C1(x) \text{ AND } \dots \text{ AND } Cm(x))} |x\rangle$. All ancillas return to $|0\rangle$.

Paper schedule: clause batches + binary AND trees + deliberate delays / ancilla recycling [Appendix C].

FULL oracle time: $T_{OC} = 4c \log_2(km/c) T_{LC}$ [Eq.(10)], including compute and uncompute.

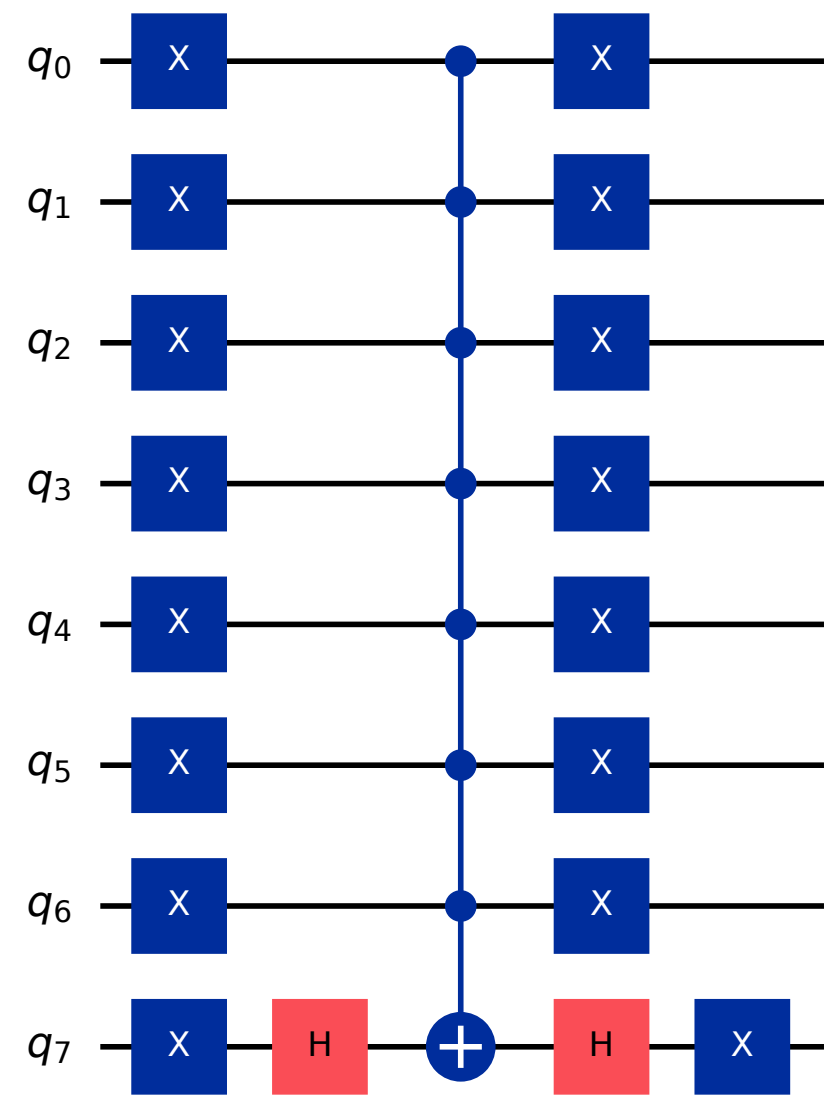
No separate paper formula allocates Eq.(10) exactly among these boxes; per-box timings are not invented.

Arrows/gates: left-to-right execution. Resource formulas belong to the paper compiler, not Qiskit circuit.depth().

Zero-state oracle: X sandwich around an all-ones phase flip

Omanakuttan et al., arXiv:2504.01897v1 (2025) | logical-circuit view; time units: T_{LC} = d x 1 microsecond

FULL ORACLE: 4 log₂(n) T_{LC} [Eq.(11)]



$O_0 = I - 2|0\dots0\rangle\langle 0\dots0| = X^n [\text{controlled phase } \pi] X^n$.

Only $|0\dots0\rangle$ receives a minus sign. $H \rightarrow MCX \rightarrow H$ implements the all-ones controlled-Z phase.

$T_{O0} = 4 \log_2(n) T_{LC}$ [Eq.(11)]; this is an approximate FULL oracle cost, not n separate costs.

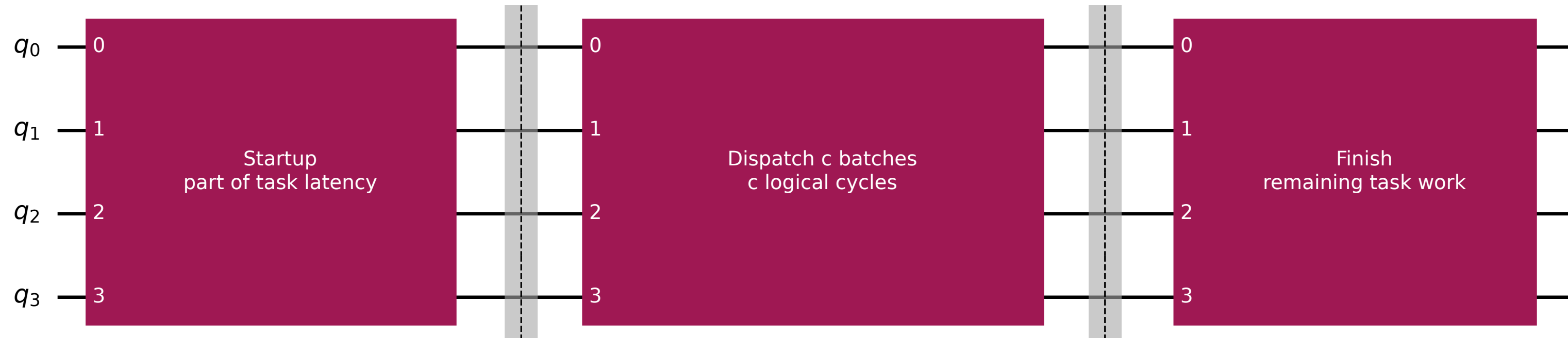
Paper implementation uses an n-qubit TACU gadget, n-1 CCZ states and logarithmic tree depth.

O_0 is used directly in Eq.(6), with no surrounding full-register H layers.

Arrows/gates: left-to-right execution. Resource formulas belong to the paper compiler, not Qiskit circuit.depth().

Phaser budget: dispatch plus startup and finish

Omanakuttan et al., arXiv:2504.01897v1 (2025) | logical-circuit view; time units: $T_{LC} = d \times 1$ microsecond



Resource-budget schematic, not a serial gate schedule or the implemented MCX circuit.

Dispatch c disjoint batches at one batch per logical cycle, with separate ancillas in flight.

Startup before first dispatch plus finish after last dispatch fit within $nP + 4 \lceil \log_2 k \rceil$.

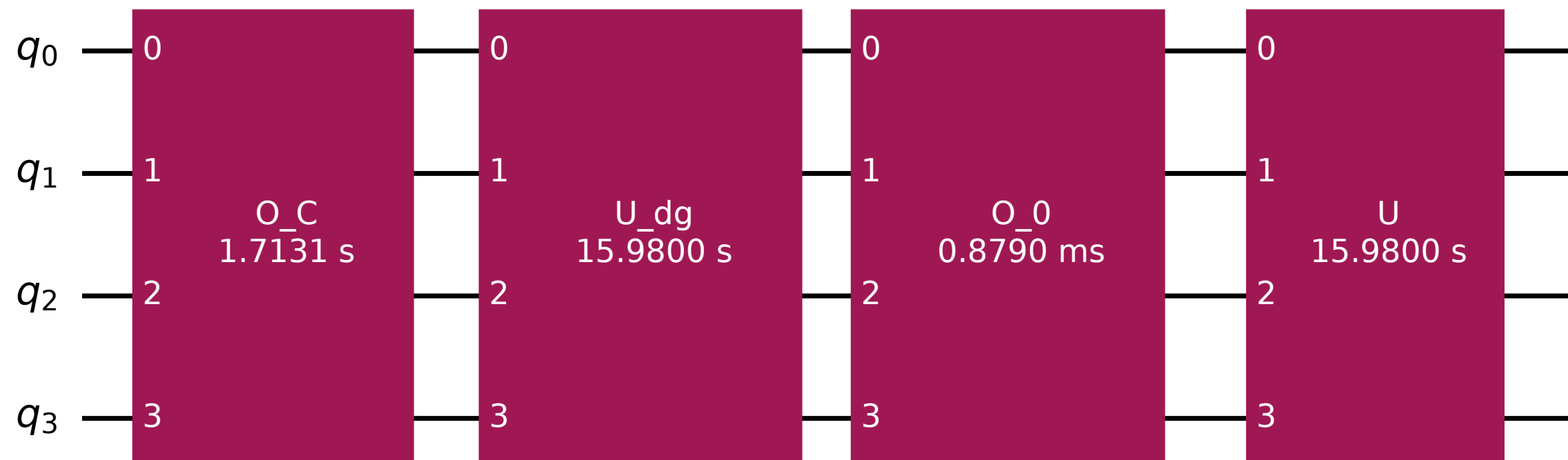
Single task: AND write $3L + \text{phase } nP + \text{measurement clear } L$; $L = \lceil \log_2 k \rceil$.

Source: p.11 Corollary IV.1 (upper bound), p.23 Appendix B; Eq.(9) uses $c + nP + 12$ for $k=8$.

Arrows/gates: left-to-right execution. Resource formulas belong to the paper compiler, not Qiskit circuit.depth().

Numerical time example for ONE AA round (not a published crossover reproduction)

n=191, m=33616, k=8, p=253, c=2112, nP=27, d=29; T_{LC}=29 microseconds



T_{mixer} = 0.783 ms [Eq.8]; T_{phaser} = 62.379 ms [Eq.9].

One U: 15.979986 s. One AA round: 33.673949 s [bracket in Eq.7].

Modeled success probability = 1.76623e-07; R = 1868.813 (continuous asymptotic estimate).

Eq.(7) AA-only: 17.4806 h; plus one initial U: 17.4851 h.

Main-text robust-AA overhead ~4x: 69.9226 h (protocol cost; Sec.II.B/IV.A).

Paper budget only; not a success/runtime guarantee for the literal Eq.(6) simulator.

c=2112 and nP=27 are selected illustrative inputs. Qiskit drawing qubits are NOT n=191 resource inputs.